MILITARY SPECIFICATION

CONTAINER, AMMUNITION, PLASTIC: PA-83

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of an ammunition container designated as Container, Ammunition, Plastic: PA-83.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification of the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9243800 - Container, Ammunition, Plastic: PA-83

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS
ASTM D 1238 - Measuring Flow Rates of Thermoplastics By Extrusion Plastometer

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.1.1 Reworked Material. No reworked material shall be used to make the product.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9243800 and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Free fall drop. The loaded (5 inert mines) container assembly shall show no evidence of broken, cracked or loose parts which render the assembly unusable, after being dropped from a height of thirty-six (36) inches (in.) onto a one (1) in. minimum (min.) thick steel plate. Slight dents where the container assembly contacted the impact surface will be acceptable, provided usability of container is not affected. Test shall be conducted at temperature extremes of minus $25^{\circ}F \pm 5^{\circ}F$ (minus $32^{\circ}C$) and $+125^{\circ}F \pm 5^{\circ}F$ ($+52^{\circ}C$).

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.5 Workmanship. All components shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, cracks, splits, cold flow, shrinkage, cold shots, inclusions, porosity, dirt, grease, surface defects, blisters, silver streaking (cap), twist or scrap, or any other defect which may affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item.

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of five (5) parts from each cavity, except ten (10) tabs (See Table 1), used to produce plastic parts and so identified as to permit tracing each item to the cavity in which it was molded .

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I: First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63403(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Parts For Container, Ammunition, Plastic: PA-83				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body, Container, Plastic (Dwg 9243801)				
	Examination for defects	5		3.2	4.4.2.1
	Cap, Plastic (Dwg 9243802)				
	Examination for defects	5		3.2	4.4.2.2
	Retainer, Tab, Plastic (Dwg 9243803)				
	Examination for defects	10		3.2	4.4.2.3
	Container, Ammunition, Plastic: PA-83 Dwg. 9243800				
	Examination for defects	5(a)		3.2	4.4.2.4
	Free fall drop test	4(a)		3.3	4.5.3

NOTES: (a) Five Containers, Ammunition, Plastic may be assembled from samples used in other first article component inspections.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Body, Container, Plastic				9243801
					NEXT HIGHER ASSEMBLY
					9243800
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside dimensions incorrect for full depth		0.40%	3.2	Gage
102	Excessive warpage of container sides		0.40%	3.2	Gage
103	Location of hole(s)		0.40%	3.2	Gage
104	Diameter of hole(s)		0.40%	3.2	Gage
105	Hole chamfer missing		0.40%	3.2	Visual
106	Outside overall length		0.40%	3.2	Gage
107	Outside width		0.40%	3.2	Gage
108	Outside height		0.40%	3.2	Gage
Minor					
201	Color improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63403(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Cap, Plastic				9243802
					NEXT HIGHER ASSEMBLY
					9243800
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside length		0.40%	3.2	Gage
102	Inside width		0.40%	3.2	Gage
103	Location of hole(s)		0.40%	3.2	Gage
104	Diameter of hole(s)		0.40%	3.2	Gage
105	Outside width		0.40%	3.2	Gage
106	Outside height		0.40%	3.2	Gage
107	Hole chamfer missing		0.40%	3.2	Visual
Minor					
201	Color improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Retainer, Tab, Plastic				9243803
					NEXT HIGHER ASSEMBLY
					9243800
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance between hemispheres		0.40%	3.2	Gage
102	Spherical radius, 2 places		0.40%	3.2	Gage
103	Height of hemisphere, 2 places		0.40%	3.2	Gage
104	Diameter of shaft, 2 places		0.40%	3.2	Gage
105	Height of shaft to hemisphere		0.40%	3.2	Gage
Minor					
201	Color improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63403(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Container, Ammunition, Plastic: PA-83				9243800
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length, max.		0.40%	3.2	Gage
102	Any component missing or damaged		0.40%	3.2	Visual
103	Difficulty in placing Cap over Container Body		0.40%	3.2	Manual
104	Removal of Tab Retainer from Container Body causes damage that prevents reclosing/reuse		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Melt Flow Rate (See dwgs. 9243801 and 9243802). Ten percent of the containers of raw material shall be selected at random, and a sample removed from each for this test. If any sample fails to comply with the applicable drawing requirement, all remaining containers of raw material shall also be sampled and tested. Any sample failing to comply with the applicable drawing requirement shall be cause for that container to be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Melt Flow Rate. The melt flow rate shall be determined in accordance with the test procedure specified in ASTM D1238. Observation shall be made for compliance with the applicable drawing requirement.

4.5.2 Dimensional control of molded or plastic parts. In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.1. through 4.4.2.3. after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for

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each minor defect. All cavities removed from production because of some fault, may, after reworking be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be submitted per 6.3 for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM , Dover, New Jersey 07801 ATTN: DRDAR-QAA-M. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5.3 Free fall drop. Four (4) container assemblies, two (2) each at the temperatures specified in 3.3, with five (5) inert mines installed in each, shall be dropped freely from the specified height on the top (cap end), one side, 45° to one corner of the top, and 45° diagonally to the opposite corner of the base (a total of four drops for each container assembly). The simulated mines shall measure 4.75 - .025 in. diameter by 2.60 - .025 in. high (right circular cylinder), and weigh 4-1/4 pounds. Observation shall be made for compliance with the applicable requirement. (Non-destructive test)

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Commercial. Unless otherwise specified by the procuring activity, the preservation and packaging shall be in accordance with MIL-STD-1188 except that the unit package must contain eight (8) plastic container assemblies.

5.2 Packing and Marking

5.2.1 Commercial. Unless otherwise specified by the procuring activity, all packing and marking shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended Use. The Plastic Ammunition Container is intended for use as the interior unit packaging for the XM74, XM75 and XM79 mines; and as a means of handling the mines during field loading operations of the XM128 Dispenser (GEMSS).

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit equipment designs as required to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading "US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)" may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the compliance of ARRADCOM.

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